

■ Sweet County Bakery

Full-Stack Artisanal E-Commerce Platform with 3D Product Spotlights & Animated Packing

Live URL: <https://sweet-county.vercel.app>

Architecture: React 19 + Node.js Express + Vercel Serverless Functions

■ Executive Summary & Business Need

Sweet County Bakery is a modern e-commerce web application engineered to transform online artisanal bakery sales through visual interactivity and reassuring user flows.

Core Project Objectives:

- **High-Fidelity Visual Appeal:** Deliver a wow-factor experience with glassmorphism, 3D perspective tilts, and dynamic bakery backgrounds.
- **Interactive Product Discovery:** Enable customers to inspect 44 artisanal treats (cakes, pastries, donuts, brownies) in mid-air floating 3D space.
- **Delightful Checkout Experience:** Implement a custom 3D bakery box packing animation before payment.
- **Zero-Delay Serverless Deployment:** Unify frontend & backend into a single-domain Vercel deployment with instant response times.

■ Technology Stack & Infrastructure

The platform is built on modern web technologies ensuring speed, security, and scalability:

Component	Technologies Used	Key Responsibilities
Frontend UI	React 19, Vite, Vanilla CSS 3D	Single Page Application, 3D canvas, state management
Routing & State	React Router v7, Context API	Client-side navigation, AuthContext, CartContext
Backend API	Node.js, Express, Vercel Serverless	Authentication, product catalog, orders, payment verification
Database	Embedded In-Memory / MongoDB Atlas	User accounts, product catalog, order history
Hosting Platform	Vercel (Single-Domain Edge)	Static asset serving & serverless API execution

■■ System Architecture & Data Flow

The application leverages an **All-in-One Serverless Architecture** hosted entirely on Vercel:

- **Unified Single Domain:** Both the React web application and Express API operate under <https://sweet-county.vercel.app>.
- **Serverless API Rewrites:** API calls to `/api/*` automatically map to Vercel Serverless Functions (`api/index.js`) with 0ms cross-origin latency.
- **SPA Client-Side Routing:** Wildcard fallback rewrite (`/(.*) -> /index.html`) prevents 404 errors during page refreshes or direct URL access.
- **Instant In-Memory Persistence:** Pre-loaded with 44 artisanal bakery items, default admin credentials, and live order tracking.

■ Feature 1: 3D Levitating Cake Spotlight

Clicking any cake card opens an interactive 3D spotlight modal designed to captivate customers:

- **Real-Time 3D Mouse Tilt:** Hovering over the spotlight tilts the cake container in 3D perspective based on cursor coordinates.
- **Smooth Mid-Air Floating Motion:** Cake image levitates gracefully in mid-air using CSS keyframes (@keyframes floatingCakeAir).
- **Pulsing Floor Shadow:** Dynamic floor shadow expands and contracts underneath the floating cake (@keyframes floatingShadowPulse).
- **Orbital Sparkles:** Decorative emojis (■ ■ ■ ■) float around the stage in 3D space.

■ Feature 2: 3D Bakery Box Packing Animation

To create excitement before payment, clicking 'Pack & Proceed to Payment' triggers a 4-step 3D packing sequence:

1. **Cake Drop (Step 1):** Selected cake drops into a 3D Sweet County Bakery Box (@keyframes dropCakeIntoBox).
2. **Lid Fold (Step 2):** Branded lid folds down to seal the box tightly.
3. **Ribbon Tie (Step 3):** Satin golden ribbon wraps across the box and ties into a bow.
4. **Wax Seal Stamp (Step 4):** Crimson wax stamp (SEALED FRESH • SWEET COUNTY) presses onto the box before transitioning to payment.

■ Feature 3: Reassuring Checkout & Payment

The payment page provides a reassuring, hassle-free ordering experience with multiple options:

- **Razorpay Online Gateway Test Modal:** Interactive checkout popup supporting UPI QR Code (GPay, PhonePe, Paytm, BHIM), Credit/Debit Cards, and Netbanking.
- **Cash on Delivery (COD):** One-click COD order booking for customers who prefer paying at their doorstep.
- **Sweet County Delight Guarantee:** Reassuring trust badge offering 100% fresh replacement guarantee.
- **Order Confirmation Screen:** Celebration screen with unique Order ID and 45-minute fresh delivery estimate.

■ Feature 4: Admin Dashboard & Customer Database

Admins have full visibility over order operations and registered customer accounts:

- **Order Pipeline Management:** Real-time status update controls (Pending → Confirmed → Out for Delivery → Delivered → Cancelled).
- **Revenue & Order Analytics:** Live tracking of total revenue, pending orders, and delivery metrics.
- **■ User Database View:** Admin tab displaying all registered customer names, emails, user IDs, and account roles.
- **Security Controls:** Change admin password and create new admin accounts directly from the dashboard.

■ Challenges, Technical Solutions & Production Deployment

Key engineering hurdles encountered during deployment and their technical solutions:

#	Challenge Encountered	Root Cause	Technical Solution
1	Could not connect to server	Frontend called localhost / sleeping Render free tiers	Unified frontend & backend into Vercel Serverless Functions
2	500 Serverless API Error	package.json specified type:module vs CommonJS exports	Converted api/index.js to ES Module format (export default handler)
3	404 Error on Page Refresh	Vercel looked for static cart.html files on refresh	Added SPA catch-all rewrite to vercel.json (/(*.*) -> /index.html)

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